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MTH301 - Calculus II (Quiz No.3)

Quiz Start Time: 04:57 PM, 27 January 2023

Question # 8 of 10 (Start time: 05:05:32 PM, 27 January 2023)

Total Marks:

For a function $\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$ in 3 - space we define $\lim\limits_{t\to a} \vec{r}(t) =$ ________.

Select the correct option

Reload Math Equations

- $x(t)\hat{i} + \left(\lim\limits_{t\to a} y(t)\right)\hat{j} + \left(\lim\limits_{t\to a} z(t)\right)\hat{k}$
- $\left(\lim\limits_{t\to a} x(t)\right)\hat{i} + \left(\lim\limits_{t\to a} y(t)\right)\hat{j}$
- $\left(\lim\limits_{t\to a} x(t)\right)\hat{i} + \left(\lim\limits_{t\to a} y(t)\right)\hat{j} + z(t)\hat{k}$
- $\left(\lim\limits_{t\to a} x(t)\right)\hat{i} + \left(\lim\limits_{t\to a} y(t)\right)\hat{j} + \left(\lim\limits_{t\to a} z(t)\right)\hat{k}$

Click to Save Answer & Move to Next Question

LIMITS OF VECTOR VALUED FUNCTIONS

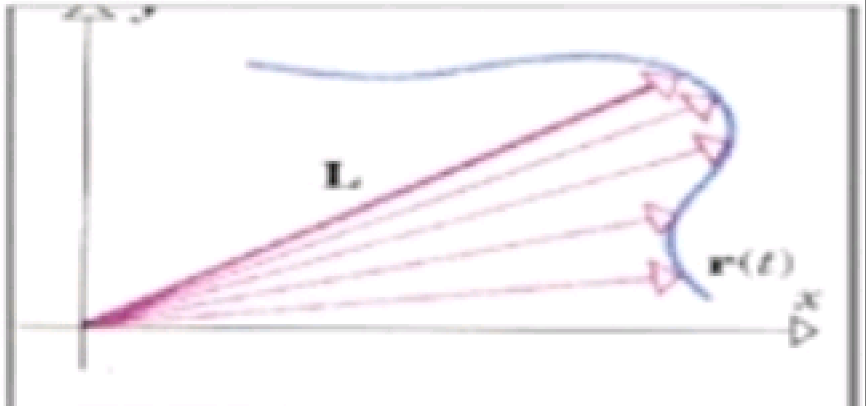
The limit of a vector-valued function is defined to be the vector that results by takin limit of each component. Thus, for a function $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$ in 2-space we defin

$$\lim_{t \to \alpha} \mathbf{r}(t) = (\lim_{t \to \alpha} x(t))\mathbf{i} + (\lim_{t \to \alpha} y(t))\mathbf{j}$$

and for a function $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$

in 3-space we define.

$$\lim_{t \to \alpha} \mathbf{r}(t) = (\lim_{t \to \alpha} x(t))\mathbf{i} + (\lim_{t \to \alpha} y(t))\mathbf{j} + (\lim_{t \to \alpha} z(t))\mathbf{k}$$



$\mathbf{r}(t)$ approaches $\mathbf{L}$ in length and direction if $\lim_{t \to a} \mathbf{r}(t) = \mathbf{L}$.

If the limit of any component does not exist, then we shall agree that the limit of $\mathbf{r}(t)$ does not exist.

These definitions are also applicable to the one-sided limits $\lim$, $\lim$ and in

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MTH301 - Calculus II (Quiz No.3)

Quiz Start Time: 04:57 PM, 27 January 2023

Question # 2 of 10 (Start time: 04:58:52 PM, 27 January 2023)

Total Marks:

P Q

The differential equation, $dz = (2x^2 - 2xy + 3)\,dx + (6y^2 - 2x^2 + 1)\,dy$, is an exact differential equation.

1 4x

Select the correct option

Reload Math Equations

- False
- True

Click to Save Answer & Move to Next Question

We now work in reverse.

Any expression $dz = Pdx + Qdy$, where P and Q are functions of x and y, is an ex
differential if it can be integrated to determine z.

$$\therefore \; P = \frac{\partial z}{\partial x} \text{ and } Q = \frac{\partial z}{\partial y}$$

Now $\dfrac{\partial P}{\partial y} = \dfrac{\partial^2 z}{\partial y \partial x}$ and $\dfrac{\partial Q}{\partial x} = \dfrac{\partial^2 z}{\partial x \partial y}$ and we know that $\dfrac{\partial^2 z}{\partial y \partial x} = \dfrac{\partial^2 z}{\partial x \partial y}$

Therefore, for dz to be an exact differential $\dfrac{\partial P}{\partial y} = \dfrac{\partial Q}{\partial x}$ and this is the test we apply.

EXAMPLE

$$dz = (3x^2 + 4y^2)\, dx + 8xy\, dy.$$

If we compare the right-hand side with $Pdx + Qdy$, then

Question # 4 of 10 (Start time: 05:02:02 PM, 27 January 2023) Total Marks:

Unless we are directed otherwise, we always proceed round the closed boundary in a (an) _________ manner.

Select the correct option

- clockwise
- anticlockwise

Click to Save Answer & Move to Next Question

$$\therefore\ I_4 = \int_2 (1+y^2)\,dy = \left[y + \frac{y^3}{3} \right]_2 = -4\frac{2}{3}$$

$$\text{Finally } I = I_1 + I_2 + I_3 + I_4 = 0 + 4\frac{2}{3} - 8 - 4\frac{2}{3} = -8 \qquad \therefore I = -8$$

Remember that, unless we are directed otherwise, we always proceed round the closed boundary in an anticlockwise manner.

Line integral with respect to arc length

We have already established that

$$I = \int_{AB} F_t\,ds = \int_{AB} \{Pdx + Qdy\}$$

where F_t denoted the tangential force along the curve c at the sample point K(x,y). The same kind of integral can, of course, relate to any function f(x,y) which is a function

MTH301 - Calculus II (Quiz No.3)

Quiz Start Time: 04:57 PM, 27 January 2023

Question # 3 of 10 (Start time: 05:00:29 PM, 27 January 2023)

Total Marks:

The graph of $r = (1+t)i + (-2+3t)j - 4tk$ is the

Select the correct option

Page # 140, 141

Reload Math Equations

- line that passes through the point (2, 1, -4)
- None of these
- line that passes through the point (1, 3, -4)
- line that passes through the point (1, -2, 0)

Click to Save Answer & Move to Next Question

Question # 1 of 10 (Start time: 04:57:17 PM, 27 January 2023)

Total Marks:

In polar coordinates, $\iint\limits_R f(x,\ y)\,dx\,dy\ = ------$

Select the correct option

Reload Math Equations

- $\iint\limits_G f(r\cos\theta,\ r\sin\theta)\,r dr\,d\theta$
- $\int\limits_G f(r\cos\theta,\ r\sin\theta)\,r dr\,d\theta$
- $\iint\limits_G f(r\cos\theta,\ r\sin\theta)\,dr\,d\theta$
- $\iint\limits_G f(r\sin\theta,\ r\cos\theta)\,r dr\,d\theta$

MTH301 - Calculus II (Quiz 3)

Quiz Start Time: 09:23 AM, 25 July 2023

Question # 10 of 10 (Start time: 09:36:40 AM, 25 July 2023)

Total Marks: 1

Path of integration parallel to ________, $dx = 0$. $\therefore\ I_C = \int_C Q\,dy$.

Select the correct option

Reload Math Equations

- y - axis
- x - axis
- $dz = 0$
- z - axis

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Question # 9 of 10 (Start time: 09:35:22 AM, 25 July 2023) Total Marks: 1

The series in the form, $f(x) = A_0 + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$; n is a positive integer, known as ______ series.

Select the correct option

Reload Math Equations

- Maclaurin

- Taylor
- Fourier ✔
- Harmonic

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Question # 8 of 10 (Start time: 09:34:08 AM, 25 July 2023)

Total Marks: 1

For a closed path in a conservative field

Select the correct option

Reload Math Equations

$$\oint_C F.dr = 0$$

$$\oint_C F.dr = 1$$

$$\oint_C F.dr = \infty$$

$$\oint_C F.dr = -1$$

Question # 7 of 10 (Start time: 09:32:42 AM, 25 July 2023) **Total Marks: 1**

Wallis sine formula when n is odd

$$\int_0^{\frac{\pi}{2}} Sin^7 x dx =$$

Select the correct option

Reload Math Equations

- (selected) 6/7. 4/5. 2/3 $\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}$
- $\frac{7}{6}\cdot\frac{5}{4}\cdot\frac{3}{2}\cdot\frac{\pi}{2}$
- $\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}\cdot\frac{\pi}{2}$
- $\frac{7}{6}\cdot\frac{5}{4}\cdot\frac{3}{2}$

Click to Save Answer & Move to Next Question

Wallis sine formula when n is odd

$$\int_0^{\frac{\pi}{2}} Cos^n x dx =$$

Select the correct option

Reload Math Equations

- $\frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdots\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$

- $\frac{n-1}{2}\cdot\frac{n-1}{2}\cdot\frac{n-1}{2}\cdot\frac{n-1}{2}\cdots\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$
- $\frac{n}{2}\cdot\frac{n-2}{2}\cdot\frac{n-4}{2}\cdot\frac{n-6}{2}\cdots\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}$
- $\frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdots\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}$

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Question # 5 of 10 (Start time: 09:30:24 AM, 25 July 2023) Total Marks: 1

Consider a vector A for which $\nabla . A = 0$ at all points, i,e. for all values of x, y, z, is called a(an) ________ .

Select the correct option

Reload Math Equations

- solenoid vector

- unit vector
- constant
- scalar

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MTH301 - Calculus II (Quiz 3) **Quiz Start Time: 09:23 AM, 25 July 2023**

Question # 3 of 10 (Start time: 09:27:37 AM, 25 July 2023) **Total Marks: 1**

If $\nabla.\vec{F}=\left(\frac{\partial}{\partial x}\hat{i}+\frac{\partial}{\partial y}\hat{j}+\frac{\partial}{\partial z}\hat{k}\right).\left(x^2\hat{i}+z\hat{j}+y\hat{k}\right)$ then $div\,\vec{F}=$ ______.

Select the correct option

Reload Math Equations

- (selected) 2x $2x$
- 0
- $2x+2$
- x^2

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Question # 2 of 10 (Start time: 09:25:48 AM, 25 July 2023) Total Marks: 1

If $Pdx + Qdy + Rdw$ is an exact differential equation then $\int_C (Pdx + Qdy + Rdw)$ is________ of the path of integration.

Select the correct option

Reload Math Equations

- dependent

- independent

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Question # 1 of 10 (Start time: 09:23:50 AM, 25 July 2023) **Total Marks: 1**

To evaluate the line integral, $\int_C V(r)\,dr$, the integrand is expressed in terms of $x,\ y,\ z$ with $d\vec{r} =$ ________.

Select the correct option

Reload Math Equations

- $dx\,\hat{i}$

- $\sqrt{dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}}$
- $dx\,\hat{i} + dy\,\hat{j}$
- $dx\,\hat{i} + dy\,\hat{j} + dz\,\hat{k}$

Click to Save Answer & Move to Next Question

sec(s)

MTH301 - Calculus II (Quiz No.3)

Quiz Start Time: 04:57 PM, 27 January 2023

Question # 10 of 10 (Start time: 05:08:34 PM, 27 January 2023)

Total Marks:

If $P = 1 + 4xy$ and $Q = 5x^2$ then ________.

Select the correct option

Reload Math Equations

- $\frac{\partial P}{\partial y} \neq \frac{\partial Q}{\partial x}$
- $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$
- $\frac{\partial Q}{\partial x} = 5x^2$
- $\frac{\partial P}{\partial y} = 1 + 4x$

Search

6:13 PM 27-Jan-23

A single curve can be represented by

Select the correct option

- Only one vector-valued function
- Infinitely many vector-valued function
- None of these
- Only two vector-valued function

Click to Save Answer & Move to Next Question

perpendicular. This is the motivation for the following useful theorem, which is applicable in both 2-space and 3-space.

THEOREM:

If **r** (t) is a vector-valued function in 2-space or 3-space and $\|\mathbf{r}(t)\|$ is constant for all t, then $\mathbf{r}(t) \cdot \mathbf{r}'(t) = 0$

Proof: That is, **r**(t) and **r**′(t) are orthogonal vectors for all t. It follows from (6) with $\mathbf{r}_1(t) = \mathbf{r}_2(t) = \mathbf{r}(t)$ that

$$\frac{d}{dt}[\mathbf{r}(t).\mathbf{r}(t)] = \mathbf{r}(t).\frac{d\mathbf{r}}{dt} + \frac{d\mathbf{r}}{dt} \cdot \mathbf{r}(t)$$

or, equivalently, $\frac{d}{dt}[\|\mathbf{r}(t)\|]^2 = 2\mathbf{r}(t) \cdot \frac{d\mathbf{r}}{dt}$

Question # 7 of 10 (Start time: 05:03:59 PM, 27 January 2023)

Total Marks:

In polar coordinate system, the equation $r = a$ represents a circle with center at - - - - - - - .

Select the correct option

Reload Math Equations

- y – axis and passes through the origin.
- Origin
- None of these.
- x – axis and passes through the origin.

Click to Save Answer & Move to Next Question

Search

6:11 PM

Question # 5 of 10 (Start time: 05:02:36 PM, 27 January 2023) Total Marks:

Integration along two distinct paths joining the same two end points .... give the same results.

Select the correct option

- not necessarily
- always

Click to Save Answer & Move to Next Question

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MTH301 - Calculus II (Quiz No.3)

Quiz Start Time: 04:57 PM, 27 January 2023

Question # 6 of 10 (Start time: 05:03:09 PM, 27 January 2023)

Total Marks:

Let $\vec{r}(t) = 2t\,\hat{i} + 3t\,\hat{j}$, then $\vec{r}'(t) = 2\,\hat{i} + 3\,\hat{j}$.

Select the correct option

Reload Math Equations

True

False

Click to Save Answer & Move to Next Question

6:08 PM

Dependence of the line integral on the path of integration

We know that integration along two separate paths joining the same two end points does not necessarily give identical results. With this in mind, let us investigate the following problem.

Example 4 : Evaluate $I = \oint_C \{3x^2y^2\,dx + 2x^3y\,dy\}$ **between O (0, 0) and A (2, 4)**

(a) along c_1 i.e. $y = x^2$

(b) along c_2 i.e. $y = 2x$

(c) along c_3 i.e. $x = 0$ from (0,0) to (0,4) and $y = 4$ from (0,4) to (2,4).

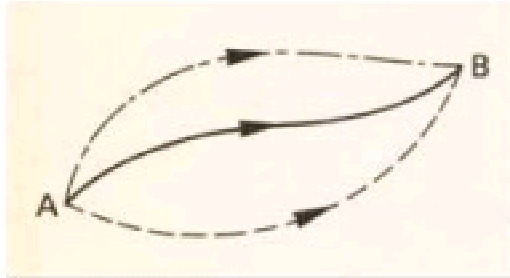
Solution:

(a).First we draw the figure and insert relevant information.

A closed path in a conservative field, $\oint_C \vec{F} . d\vec{r} =$ __________.
-1
1
2
0

Conservative Vector Fields

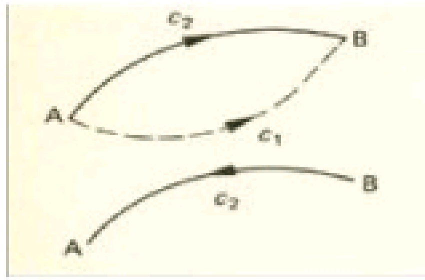
In general, the value of the integral $\int_C \mathbf{F.dr}$ between two stated points A and B depends on the particular path of integration followed. If, however, the line integral between A and B is independent of the path of integration between the two end points, then the vector field F is said to be conservative.



It follows that, for a closed path in a conservative field, $\oint_C \mathbf{F.dr} = 0$.

For, if the field is conservative,

$\int_{C_1(AB)} \mathbf{F.dr} = \int_{c_2(AB)} \mathbf{F.dr}$ But $\int_{c_2(BA)} \mathbf{F.dr} - \int_{c_2(AB)} \mathbf{F.dr}$



Hence, for the closed path $AB_{c_1} + BA_{c_2}$, $\oint \mathbf{F.dr}$

$= \int_{C_1(AB)} \mathbf{F.dr} + \int_{c_2(AB)} \mathbf{F.dr}$

$= \int_{C_1(AB)} \mathbf{F.dr} - \int_{c_2(AB)} \mathbf{F.dr} = \int_{C_1(AB)} \mathbf{F.dr} - \int_{c_1(AB)} \mathbf{F.dr} = 0$

MTH301 - Calculus II (Quiz No.3.)

Question # 2 of 10 (Start time: 10:41:55 AM, 05 July 2024)

Wallis sine formula when n is even

$$\int_0^{\frac{\pi}{2}} Sin^n x dx =$$

Select the correct option

- $\frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdots\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}$
- $\frac{n}{2}\cdot\frac{n-2}{2}\cdot\frac{n-4}{2}\cdot\frac{n-6}{2}\cdots\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}$
- $\frac{n-1}{2}\cdot\frac{n-1}{2}\cdot\frac{n-1}{2}\cdot\frac{n-1}{2}\cdots\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$
- $\frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdots\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$

Wallis Sine Formula

When n is even $$\int_0^{\frac{\pi}{2}} \sin^n x dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdot---------\frac{5}{6}\cdot\frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$$

When n is odd $$\int_0^{\frac{\pi}{2}} \cos^n x dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdot---------\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}.$$

$\therefore$ curl $\mathbf{F} = 0$

So three tests can be applied to determine whether or not a vector field is conservative. They are

(a) $\oint \mathbf{F.dr} = 0$

(b) curl $\mathbf{F} = 0$

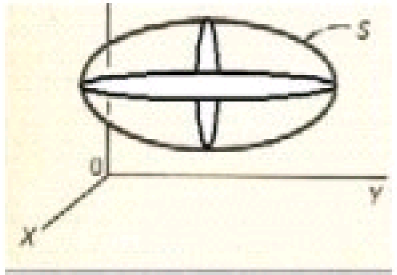
(c) $\mathbf{F}$ = grad V

Any one of these conditions can be applied as is convenient.

Divergence Theorem (Gauss' theorem)

For a closed surface S, enclosing a region V in a vector field F

$$\int_V \text{div}\, \mathbf{F}\, dV = \int_S \mathbf{F.dS}$$



In general, this means that the volume integral (tripple integral) on the left-hand side can be expressed as a surface integral (double integral) on the right-hand side.

Example

MTH301 - Calculus II (Quiz No.3.)

Question # 7 of 10 (Start time: 10:46:39 AM, 05 July 2024)

We can apply the following test/tests: ________ to determine whether or not a vector field is conservative.

Select the correct option

(d) All (a), (b) and (c).

(b) curl $\vec{F} = 0$

(c) $\vec{F} = grad\ \vec{V}$

(a) $\oint \vec{F}.\, d\vec{r} = 0$

Properties of line integrals

1. $$\int_C F\,ds = \int_C \{P\,dx + Q\,dy\}$$

2. $$\int_{AB} F\,ds = -\int_{BA} F\,ds \quad \text{and} \quad \int_{AB} \{P\,dx+Q\,dy\} = \int_{BA} \{P\,dx+Q\,dy\}$$

i.e. the sign of a line integral is reversed when the direction of the integration along the path is reversed.

3. (a) For a path of integration parallel to the y-axis, i.e. $x = k$, $dx = 0$

$$\therefore \int_C P\,dx = 0 \quad \therefore\ I_C = \int_C Q\,dy.$$

(b) For a path of integration parallel to the x-axis, i.e. $y = k$, $dy = 0$.

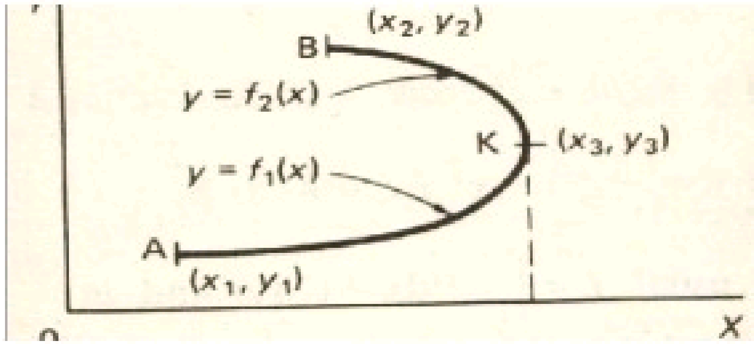
$$\therefore \int_C Q\,dy = 0 \ \therefore\ I_C = \int_C P\,dx.$$

4. If the path of integration c joining A to B is divided into two parts AK and KB, then

$$I_c = I_{AB} = I_{AK} + I_{KB}.$$

5. If the path of integration c is not single valued for part of its extent, the path is divided into two sections.

$y = f_1(x)$ from A to K, $y = f_2(x)$ from K to B.



6. In all cases, the actual path of integration involved must be continuous and single-valued.

MTH301 - Calculus II (Quiz No.3.)

Question # 1 of 10 (Start time: 10:41:01 AM, 05 July 2024)

Path of integration parallel to ________, $dy = 0$. $\therefore\ I_C = \int_C P\,dx$.

Select the correct option

○	z - axis
◉	x - axis
○	y - axis
○	$dz = 0$

Question # 8 of 10 (Start time: 10:47:30 AM, 05 July 2024)

Path of integration parallel to ________, $dx = 0$. $\therefore\ I_C = \int_C Q\,dy.$

Select the correct option

- z - axis
- $dz = 0$
- y - axis
- x - axis

MTH301 - Calculus II (Quiz No.3.)

Question # 9 of 10 (Start time: 10:48:26 AM, 05 July 2024)

If $Pdx + Qdy + Rdw$ is an exact differential equation then ______.

Select the correct option

◉	(d) All (a), (b) and (c). ✓
○	(b) $\dfrac{\partial P}{\partial w} = \dfrac{\partial R}{\partial x}$
○	(a) $\dfrac{\partial P}{\partial y} = \dfrac{\partial Q}{\partial x}$
○	(c) $\dfrac{\partial R}{\partial y} = \dfrac{\partial Q}{\partial w}$

Exact differentials in three independent variables

A line integral in space naturally involves three independent variables, but the method is very much like that for two independent variables.

$dz = Pdx + Q\,dy + R\,dw$ is an exact differential of $z = f(x, y, w)$

$$\text{if } \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x};\ \frac{\partial P}{\partial w} = \frac{\partial R}{\partial x};\ \frac{\partial R}{\partial y} = \frac{\partial Q}{\partial w}$$

If the test is successful, then

(a) $\int_C (P\,dx + Q\,dy + R\,dw)$ is independent of the path of integration.

(b) $\oint_C (P\,dx + Q\,dy + R\,dw)$ is zero.

MTH301 - Calculus II (Quiz No.3.)

Question # 6 of 10 (Start time: 10:45:00 AM, 05 July 2024)

Evaluate $\int lnx dx = --------$

Select the correct option

- ( ) $lnx - x + x$
- (•) $xlnx - x + c$
- ( ) $xlnx - \frac{1}{x} + c$
- ( ) $lnx - \frac{1}{x} + c$

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Question # 4 of 10 (Start time: 09:28:55 AM, 25 July 2023) Total Marks: 1

Line integral is used to calculate ---------

Select the correct option

- length
- volume
- area
- force

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Integration By Parts

$$\int UV dx = U\int V\, dx - \int\left[\int V dx.\frac{dU}{dx}\right]dx$$

Example *Evaluate* $\int$ x lnx dx

$\int$ x lnx dx= lnx $\int$ xdx – $\int$ [$\int$ x dx. $\frac{d}{dx}$(ln x)] dx (We are integrating by parts)

$$= \ln x(\frac{x^2}{2}) - \int (\frac{x^2}{2})(\frac{1}{x})dx = (\frac{x^2}{2})\ln x - \int (\frac{x}{2})d\,x = (\frac{x^2}{2})\ln x - \frac{1}{2}\ (\frac{x^2}{2})$$

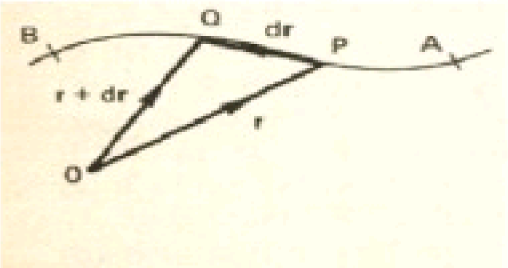
Example *Evaluate* $\int$ x sinx dx

$\int$ **x** sinx dx =x$\int$ sin xdx – $\int$ [$\int$ sinx dx. $\frac{d}{dx}$(x)] dx (We are integrating by parts)

$$= x(-\cos x) - \int (-\cos x)(1)dx = -x(\cos x) + \int \cos x\, d\,x = -x(\cos x) + \sin x$$

Line Integrals

Let a point p on the curve c joining A and B be denoted by the position vector **r** with respect to origin O. If q is a neighboring point on the curve with position vector



r+dr , then $\overline{PQ}$ = **r**

(a) Grad operator ∇ acts on a scalar field to give a vector field.

(b) Div operator $\nabla.$ Acts o a vector field to give a scalar field.

(c) Curl operator $\nabla \times$ acts on a vector field to give a vector field.

(d) With a scalar function ϕ (x,y,z)

$$\text{Grad } \phi = \nabla\phi = \frac{\partial\phi}{\partial x}\mathbf{i} + \frac{\partial\phi}{\partial y}\mathbf{j} + \frac{\partial\phi}{\partial z}\mathbf{k}$$

(e) With a vector function $A = a_x i + a_y \mathbf{j} + a_z \mathbf{k}$

$$\text{(i) div } \mathbf{A} = \nabla.\mathbf{A} = \frac{\partial a_x}{\partial x} + \frac{\partial a_y}{\partial y} + \frac{\partial a_z}{\partial z}$$

$$\text{(ii) Curl } \mathbf{A} = \nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_x & a_y & a_z \end{vmatrix}$$

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